



DSA ACTIVITY 2: Mathematics, Algorithms and Problem-Solving

Instructions:

- Write **ALL answers and solutions by hand on a piece of paper.**
- Make sure your **name, section, and date** are written at the top of your paper.
- **Show your complete solutions**, not just the final answers.
- Take a **clear and readable photo** of your completed activity before submitting.
- **Combine/organize all photos into ONE PDF file** before submitting.
- Submit the **PDF file** through our GCLamp folder with a filename **LN.FN.Activity1.**
- **Failure to follow the instructions will result in NO SCORE for the activity.**

PART A – BASIC MATHEMATICAL FUNCTIONS — 40 POINTS

A. Find the absolute value of each number. (1pt each)

1. $|-15| = \underline{\hspace{2cm}}$
2. $|24| = \underline{\hspace{2cm}}$
3. $|-36| = \underline{\hspace{2cm}}$
4. $|-8| = \underline{\hspace{2cm}}$
5. $|0| = \underline{\hspace{2cm}}$

B. Round each number to the nearest whole number. (1pt each)

1. $12.4 = \underline{\hspace{2cm}}$
2. $18.7 = \underline{\hspace{2cm}}$
3. $25.5 = \underline{\hspace{2cm}}$
4. $43.2 = \underline{\hspace{2cm}}$
5. $96.8 = \underline{\hspace{2cm}}$

C. Determine the floor and ceiling of each number. (2pts each)

Number	Floor	Ceiling
4.6		
7.2		
10.8		
15.3		
21.9		





GORDON COLLEGE COLLEGE OF COMPUTER STUDIES

OLONGAPO CITY

BS in Computer Science • BS in Entertainment and Multimedia Computing - Digital Animation Technology
BS in Entertainment and Multimedia Computing - Game Development • BS in Information Technology

D. For each pair, identify the minimum and maximum value. (2pts each)

1. 15 and 28

Minimum: _____ Maximum: _____

2. -12 and -7

Minimum: _____ Maximum: _____

3. 45 and 32

Minimum: _____ Maximum: _____

4. -25 and -40

Minimum: _____ Maximum: _____

5. 100 and 75

Minimum: _____ Maximum: _____

PART B – MODULO AND ROOT FUNCTIONS — 30 POINTS

A. Find the remainder of each operation. (2pts each)

1. $10 \% 3 =$ _____

2. $15 \% 4 =$ _____

3. $20 \% 6 =$ _____

4. $27 \% 5 =$ _____

5. $35 \% 8 =$ _____

B. Find the value of each square root. (2pts each)

1. $\sqrt{16} =$ _____

2. $\sqrt{25} =$ _____

3. $\sqrt{36} =$ _____

4. $\sqrt{49} =$ _____

5. $\sqrt{64} =$ _____

C. Find the value of each cube root. (2pts each)

1. $\sqrt[3]{8} =$ _____

2. $\sqrt[3]{27} =$ _____

3. $\sqrt[3]{64} =$ _____

4. $\sqrt[3]{125} =$ _____

5. $\sqrt[3]{216} =$ _____

PART C – EXPONENTIAL FUNCTIONS, LAWS OF EXPONENTS & POW() — 30 POINTS

A. Given: $f(x) = 2^x$. Evaluate the following. (2pts each)

1. $f(0) =$ _____

2. $f(1) =$ _____

3. $f(2) =$ _____

4. $f(3) =$ _____

5. $f(4) =$ _____

B. Simplify each expression. (2pts each)

1. $2^3 \times 2^2 =$ _____

2. $5^4 \div 5^2 =$ _____

3. $3^2 \times 3^3 =$ _____

4. $10^5 \div 10^2 =$ _____

5. $(2^3)^2 =$ _____

C. Evaluate each expression. (2pts each)

1. $\text{pow}(2, 3) =$ _____

2. $\text{pow}(3, 2) =$ _____

3. $\text{pow}(5, 2) =$ _____

4. $\text{pow}(2, 5) =$ _____

5. $\text{pow}(10, 2) =$ _____





GORDON COLLEGE COLLEGE OF COMPUTER STUDIES

OLONGAPO CITY

• BS in Computer Science • BS in Entertainment and Multimedia Computing - Digital Animation Technology
• BS in Entertainment and Multimedia Computing - Game Development • BS in Information Technology

POINT DISTRIBUTION:

Part	Topic	Points
Part A	Basic Mathematical Functions	40
Part B	Modulo and Root Functions	30
Part C	Exponential Functions, Laws of Exponents & pow()	30

TOTAL: 100 PTS



WURI

Awards Asia 2025

THE World University Rankings 2026

THE World University Rankings 2026 Asia

Times Higher Education Sustainability Impact Ratings 2026



(047) 222-4080 loc. 319

info.ccs@gordoncollege.edu.ph

www.gordoncollegeccs.edu.ph

Gordon College CCS